

10 (a)	(i)	1. inverting (amplifier)	B1	[1]
		2. gain of op-amp is very large / infinite	B1	
		non-inverting input is at earth / 0V	B1	
		for amplifier not to saturate, P must be at about earth / 0V	B1	[3]
	(ii)	input resistance is very large	B1	
		(so) current in R_1 = current in R_2	B1	
		$I = V_{IN} / R_1$	B1	
		$I = -V_{OUT} / R_2$ (minus sign can be in either of the equations)	B1	
		hence <i>gain</i> = $V_{OUT} / V_{IN} = -R_2 / R_1$	A0	[4]
(b)	(i)	1. feedback resistance = 33.3 k Ω	C1	
		gain (= 33.3 / 5) = 6.66	C1	
		V_{OUT} (= 6.66 $\times$ 1.2) = 8.0 V (+ or – acceptable, allow 1 s.f.)	A1	[3]
		2. feedback resistance = 8.33 k Ω	C1	
		V_{OUT} (= {6.66 $\times$ 1.2} / 5) = 2.0 V (+ or – acceptable, allow 1 s.f.)	A1	[2]
	(ii)	(Increase in lamp-LDR distance gives) decrease in intensity	M1	
		<u>Feedback</u> / <u>LDR</u> resistance increases	M1	
		voltmeter reading increases / becomes more negative	A1	[3]
11 (a)		GT image: (thin) slice (through structure)	B1	